

An order criterion for multilinear verbal subgroups

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This is not a preprint. This note advertises the result and explains its proof so that human researchers can check, develop and build on it. It is an agent-written mathematical account, not a refereed article.

1 The question and the answer

A multilinear commutator word, also called an outer commutator word, is obtained by repeatedly taking commutators of expressions in disjoint variables. Thus

$$[x_1, x_2], \quad [[x_1, x_2], x_3], \quad [[x_1, x_2], [x_3, x_4]]$$

are examples. We use $[a, b] = a^{-1}b^{-1}ab$, and include the one-variable word. Let G_w denote the set of individual w -values in G , and put $w(G) = \langle G_w \rangle$.

Problem 21.35, proposed by Yerko Contreras Rojas, Valentina Grazian and Carmine Monetta [1], asks whether an order condition on these values implies that $w(G)$ is p -nilpotent. Here a finite group is p -nilpotent if it has a normal subgroup of order prime to p and index a power of p , called a normal p -complement. The answer is affirmative.

Theorem 1. *Let G be a finite group, p a prime, and w a multilinear commutator word. Then $w(G)$ has a normal p -complement if and only if*

$$x, y \in G_w, \quad p \nmid |x|, \quad p \mid |y| \implies p \mid |xy|. \quad (\text{P})$$

The element y may have mixed order. The hypothesis concerns single word values, not arbitrary elements of the verbal subgroup. Contreras Rojas, Grazian and Monetta proved the lower-central-word case for arbitrary finite groups and the derived-word case for soluble groups [2, Theorems C and D].

We give an outline of the proof, including the centralization calculation and the final quasisimple argument. The soluble word calculus and the calculations for extensions of the minimal simple groups are summarized; their full proofs are longer than this note.

2 Structural results used

Thompson's classification [3] says that a finite nonabelian simple group whose proper subgroups are soluble is isomorphic to one of

$$\begin{aligned} & \text{PSL}_2(2^f), \quad f \text{ prime}; \\ & \text{PSL}_2(3^f), \quad f \text{ an odd prime}; \\ & \text{PSL}_2(\ell), \quad \ell > 3 \text{ prime}, \quad 5 \mid \ell^2 + 1; \\ & \text{Sz}(2^{2m+1}), \quad m > 0, \quad 2m + 1 \text{ prime}; \\ & \text{PSL}_3(3). \end{aligned}$$

The second result concerns a finite *quasisimple* group H : this means that $H = H'$ and $H/Z(H)$ is nonabelian simple. The theorem of Liebeck, O'Brien, Shalev and Tiep [4, Theorem 1 and Table 1] has the following consequence:

$$\gcd(|x|, |Z(H)|) = 1 \implies x = [a, b] \text{ for some } a, b \in H. \quad (\text{Q})$$

Indeed, every exception in their list has order sharing a prime divisor with the order of the center. The list incorporates Blau's work on central elements [13]. When $Z(H) = 1$, (Q) gives the authors' solution of Ore's conjecture [5]: every element of a finite nonabelian simple group is a commutator.

We also use Frobenius' normal p -complement criterion: a finite group H is p -nilpotent if $N_H(P)/C_H(P)$ is a p -group for every p -subgroup P of H ; see [6, Theorem 5.26].

3 From orders to centralization

Write $\delta_0 = x_1$ and $\delta_{k+1} = [\delta_k, \delta_k]$, with disjoint variables in the two copies. If w has height h , completing its binary tree shows that every δ_h -value is a w -value. Condition (P) therefore holds for δ_k whenever $k \geq h$.

The following lemma is the centralization argument of Contreras Rojas, Grazian and Monetta [2, Lemmas 5.1 and 5.2].

Lemma 2. *Suppose (P) holds for δ_k . If a δ_k -value x has order prime to p and normalizes a p -subgroup P , then x centralizes P .*

Proof. Take $g \in P$ and set $c = [g, x]$ and $t = [c, x]$. The identity

$$t = x^{-1}[g^{-1}x^{-1}g, x]x$$

expresses t as a conjugate of a δ_{k+1} -value, hence as a single δ_k -value. Here we use closure of derived-word values under inversion and conjugation. Also x^{-1} is such a value, and

$$x^{-1}t = (cx)^{-1}x^{-1}(cx).$$

If $p \mid |t|$, (P) would imply $p \mid |x^{-1}t|$, contrary to this conjugacy. But $c, t \in P$, so $t = 1$. Hence c commutes with x . Since

$$cx^{-1} = g^{-1}x^{-1}g,$$

raising to the power $|x|$ gives $c^{|x|} = 1$. The order of c is a power of p , and therefore $c = 1$. This holds for every $g \in P$. $\square$

4 The least-counterexample reduction

First suppose that G is soluble. One proves the theorem by induction on the defect of the tree of w , meaning the number of vertices missing from the complete binary tree of the same height. Defect zero is the derived-word case of [2]. A proper extension of the tree of the same height decreases the defect, and its values remain w -values.

Pass to $\bar{G} = G/O_{p'}(G)$; condition (P) descends through this quotient. The induction hypothesis puts the verbal subgroups of the proper extensions inside $O_p(\bar{G})$. The outer-word commutator

calculus then shows that each p' -order w -value centralizes $O_p(\overline{G})$. This is the Fitting subgroup of $\overline{G}$ and is self-centralizing, so such a value is trivial. The soluble verbal-generation argument gives that $w(\overline{G})$ is a p -group. Lifting through the normal p' -kernel proves the soluble case. The word-tree, section and generation methods here use the results of [7], [8], [9] and [10].

Fix w, p , and suppose now that G is a counterexample of least order. Write R for its soluble radical. Minimality gives

$$\begin{aligned} G &= G', & O_{p'}(G) &= 1, & R &= \Phi(G) = O_p(G), \\ S &= G/R \text{ nonabelian simple,} \end{aligned} \tag{1}$$

with $p \mid |S|$. Here $\Phi(G)$ is the Frattini subgroup.

To explain these reductions, a normal p -complement lifts through a normal p' -kernel, giving $O_{p'}(G) = 1$. For a proper normal subgroup H , minimality gives a normal p -complement in $w(H)$. This complement is characteristic in $w(H)$ and hence normal in G , so it is trivial. Thus $w(H)$ is a p -group. Since $H^{(h)} \leq w(H)$, the subgroup H is soluble. In particular G is perfect and G/R is nonabelian simple.

A proper supplement to R would contain a perfect supplement K . Minimality makes K a p' -group, and the centralization lemma makes it centralize $O_p(G)$. The normal-centralizer argument then forces R to be central, whereas a central radical with a perfect supplement gives $G' = K < G$, a contradiction. Hence $R \leq \Phi(G)$. The reverse containment, nilpotence of the finite Frattini subgroup and $O_{p'}(G) = 1$ give the equality in (1). Finally, the coprime complement theorem excludes $p \nmid |S|$.

The principal intermediate result is the following.

Proposition 3. *A least counterexample satisfying (1) has p odd and $R \leq Z(G)$.*

We describe the two parts of its proof. Suppose first that R is noncentral. Then $C_G(R)$ is a proper normal subgroup, hence soluble and contained in R . By the centralization lemma, every p' -order δ_k -value, for $k \geq h$, lies in $C_G(R) \leq R$. It must therefore be trivial.

If $H < G$ is perfect, then $w(H) = H$, and minimality gives a normal p -complement in H . The quotient by this complement is both perfect and a p -group, hence trivial; thus H has p' -order. Its δ_k -values all vanish, forcing $H = 1$. Applying this to the terminal derived subgroup of each proper subgroup of G shows that every proper subgroup is soluble. The same holds for S , so Thompson's list applies.

For odd p , the contradiction comes from a nonidentity 2-element that is a value of every outer word. A *generating good set* in a group X is a subset B such that each $t \in B$ has an expression $t = [a, b]$ with $a, b \in B$ and $\langle a, b \rangle = X$. Recursion down a word tree makes every member of B a single value of every outer word.

In each of Thompson's groups one constructs such a set B , a 2-subgroup U , and elements $d \in B \cap N_S(U)$ and $u \in U$ with

$$[d, u^{-1}du] \neq 1. \tag{2}$$

The even-field projective linear and Suzuki cases use root 2-subgroups and split tori; the odd-field projective linear cases use a four-group with a tetrahedral normalizer. The $\text{PSL}_3(3)$ case uses a semidihedral 2-subgroup and an order-four good class. The split-torus calculations include Brandl's construction [11, Lemma 4(c)].

The lifting argument passes through the central quotient $G/[R, G]$. A suitable good set there maps onto B , and its full inverse image is good in G . Combining this with the coprime normalizer argument gives a lift a of d normalizing a 2-subgroup P that maps isomorphically onto U , while retaining a as a value of every outer word. Lift u to $b \in P$. Then

$$z = [a, b^{-1}ab] \in P$$

has nontrivial image by (2). It is a value of every nonleaf outer word: use a as a value of the left child and its conjugate as a value of the right child. The leaf case is immediate. Since p is odd, z has p' -order, contradicting the vanishing of the δ_k -values. This excludes a noncentral radical when p is odd.

The prime 2 requires a separate argument. First, Thompson's theorem implies solubility of every odd-order group: a least nonsoluble odd-order group would be minimal nonabelian simple, whereas every group on the list has even order. A proper perfect subgroup of a least counterexample at $p = 2$ has odd order, by the normal-complement argument above, and is therefore trivial. Thus S is minimal simple even when R is central.

One must now exclude perfect Frattini 2-extensions of the five families. This requires control of commutator fibres through the 2-kernel, using linear algebra and representation theory. The even-field and odd-field projective linear cases, $\text{PSL}_3(3)$ and the Suzuki cases yield the required order obstruction. The smallest parameters, including $\text{PSL}_2(4)$ and $\text{Sz}(8)$, are treated separately. The representation calculations include the relevant characteristic-two instance of Steinberg's tensor description [12]. These extension calculations are a substantial part of the proof and are not reproduced here. Together with the odd-prime argument, they establish the reduction proposition.

5 The central radical and the conclusion

It remains to exclude $R \leq Z(G)$. Since G/R has trivial center, (1) gives

$$R = Z(G) = O_p(G).$$

Thus G is quasisimple with p -power center. By (Q), every p' -element of G is an ordinary commutator. The following observation converts these elements into values of the required word.

Lemma 4. *Let $\pi : H \twoheadrightarrow S$ have central kernel, and suppose every element of S is a commutator. Then every nonleaf outer commutator word has the same set of values in H as the ordinary commutator word.*

Proof. Induction on the word shows that every outer word is surjective on S . Given $v = [v_1, v_2]$ and an ordinary commutator $[a, b] \in H$, choose a v_1 -value a' and a v_2 -value b' whose images are $\pi(a)$ and $\pi(b)$. Such lifts exist because word evaluation commutes with a surjective homomorphism. The elements $a'a^{-1}$ and $b'b^{-1}$ are central, so

$$[a', b'] = [a, b].$$

Hence every ordinary commutator is a v -value. The reverse inclusion follows from the outermost bracket. $\square$

Apply the lemma to $G \twoheadrightarrow G/Z(G)$, using Ore's theorem. Every p' -element of G is consequently a value of every outer word, including δ_h ; the one-variable case is immediate. For any p -subgroup

P , the centralization lemma now shows that every p' -element of $N_G(P)$ centralizes P . Each Sylow q -subgroup of $N_G(P)$, for $q \neq p$, is therefore contained in $C_G(P)$, and $N_G(P)/C_G(P)$ is a p -group. Frobenius' criterion gives a normal p -complement in G , and hence in its subgroup $w(G)$, a contradiction. This completes the forward implication, with the intermediate calculations summarized above.

Conversely, suppose $K \triangleleft w(G)$ is a normal p -complement. Every p' -element $x \in w(G)$ lies in K . If $y \in w(G)$ and $p \mid |y|$, then $y \notin K$, so its image in the p -group $w(G)/K$ is nonidentity. The element xy has the same image, giving $p \mid |xy|$. In particular, (P) holds. $\square$

6 Credit and the formal record

The question, the earlier cases and the centralization argument are due to Contreras Rojas, Grazian and Monetta. The classification, commutator, word-calculus and representation-theoretic results cited above are prior work. Nilradical contributed the additional word and extension arguments and their assembly into the general result, together with its formalization.

The Lean formalization takes exactly two mathematical hypotheses: Thompson's minimal-simple classification and (Q). Their published proofs are not formalized here. All other steps, including Frobenius' criterion, are proved dependencies. We acknowledge Yawara Ishida and the OddOrder contributors, the Qiuzhen-CFSG and Tau Ceti contributors, and the Lean and mathlib communities for reused mathematics and code; Comparator, lean4export and Nanoda for verification tools; and The GAP Group for computational software.

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